

SHETH L.U.J AND SIR M.V COLLEGE

PRACTICAL NO: 15

Aim :- Basic MANET Implementation with Packet Animation

Simulate a mobile ad-hoc network and visualize packet movement to study dynamic topology formation.

Manet.tcl

```
# =====
# Basic MANET Simulation with Packet Animation
# =====

set val(chan)    Channel/WirelessChannel
set val(prop)    Propagation/TwoRayGround
set val(netif)   Phy/WirelessPhy
set val(mac)     Mac/802_11
set val(ifq)     Queue/DropTail/PriQueue
set val(ll)      LL
set val(ant)     Antenna/OmniAntenna
set val(ifqlen)  50
set val(nn)      5
set val(rp)      AODV
set val(x)       500
set val(y)       500
set val(stop)    30

# Simulator
set ns [new Simulator]

# Trace Files
set tracefd [open manet.tr w]
$ns trace-all $tracefd

set namfd [open manet.nam w]
$ns namtrace-all-wireless $namfd $val(x) $val(y)

# Topography
set topo [new Topography]
$topo load_flatgrid $val(x) $val(y)

# GOD Object
create-god $val(nn)

# Channel
set chan_1_ [new $val(chan)]

# Wireless Configuration
$ns node-config \
```

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```
-adhocRouting $val(rp) \  
-llType $val(ll) \  
-macType $val(mac) \  
-ifqType $val(ifq) \  
-ifqLen $val(ifqlen) \  
  
-antType $val(ant) \  
-propType $val(prop) \  
-phyType $val(netif) \  
-channel $chan_1_ \  
-topoInstance $topo \  
-agentTrace ON \  
-routerTrace ON \  
-macTrace ON \  
-movementTrace ON  
  
# Create Nodes  
for {set i 0} {$i < $val(nn)} {incr i} {  
    set node_  
i)[ns node]  
}  
  
# Initial Positions  
$node_(0) set X_ 50  
$node_(0) set Y_ 50  
$node_(0) set Z_ 0  
  
$node_(1) set X_ 150  
$node_(1) set Y_ 150  
$node_(1) set Z_ 0  
  
$node_(2) set X_ 250  
$node_(2) set Y_ 250  
$node_(2) set Z_ 0  
  
$node_(3) set X_ 350  
$node_(3) set Y_ 350  
$node_(3) set Z_ 0  
  
$node_(4) set X_ 450  
$node_(4) set Y_ 450  
$node_(4) set Z_ 0  
  
# Display Nodes in NAM  
for {set i 0} {$i < $val(nn)} {incr i} {  
    node_  
i) random-motion 0  
    $ns initial_node_pos node_  
i) 20
```

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```
}

# Mobility
nsat2.0"node_(0) setdest 400 100 15"
nsat4.0"node_(1) setdest 450 250 10"
nsat6.0"node_(2) setdest 100 400 12"

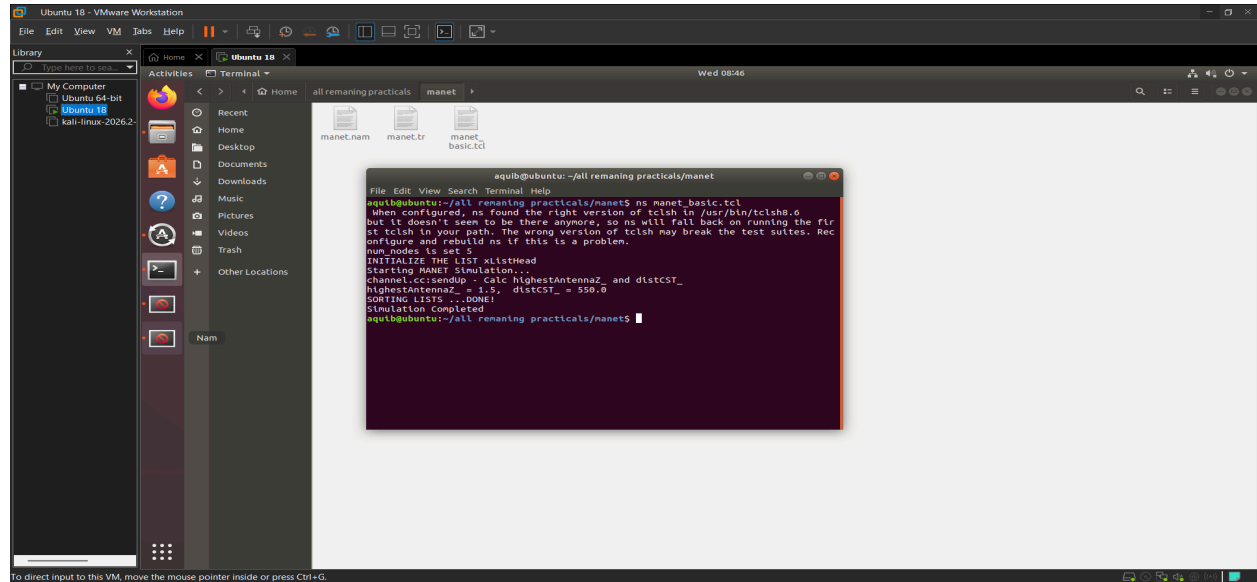
# UDP Agent
set udp0 [new Agent/UDP]
$ns attach-agent $node_(0) $udp0

set null0 [new Agent/Null]
$ns attach-agent $node_(4) $null0
$ns connect $udp0 $null0
# CBR Traffic
set cbr0 [new Application/Traffic/CBR]
$cbr0 set packetSize_ 512
$cbr0 set rate_ 200k
$cbr0 attach-agent $udp0
nsat5.0"cbr0 start"
nsat25.0"cbr0 stop"
# End NAM
$ns at $val(stop)"ns nam-end-wireless $val(stop)"
# Finish Procedure
proc finish {} {
    global ns tracefd namfd
    $ns flush-trace
    close $tracefd
    close $namfd
    puts "Simulation Completed"
    exec nam manet.nam &
    exit 0
}
$ns at $val(stop) "finish"
puts "Starting MANET Simulation..."
$ns run
```

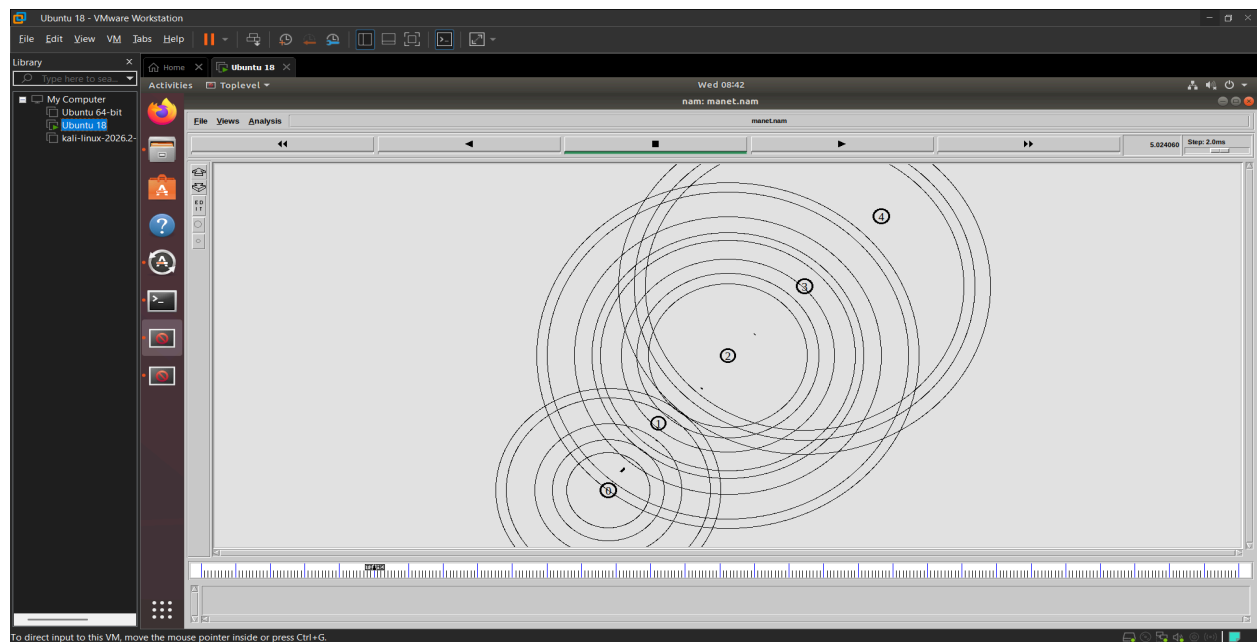
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OUTPUT:-



```
File Edit View Search Terminal Help
aquib@ubuntu:~/all remaining practicals/manet$ ns manet_basic.tcl
When configured, ns found the right version of tclsh in /usr/bin/tclsh8.6
but it doesn't seem to be there anymore, so ns will fall back on running the fir
st tclsh in your path. The wrong version of tclsh may break the test suites. Rec
onfigure and rebuild ns if this is a problem.
num_nodes is set 2
INITIALIZE THE LIST xListHead
Starting MANET Simulation...
channel.ccisendup - calc_highestAntennaZ_ and distCST_
highestAntennaZ_ = 1.5, distCST_ = 550.0
SORTING LISTS ...DONE!
Simulation Completed
aquib@ubuntu:~/all remaining practicals/manet$
```



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